

Item	Operating VDC	Current Consumption	SNR	Quantity	Power(W) = V X I
Cytron MD20A	6-30V Motor Supply; logic accepts 1.5-15V HIGH	Per channel up to 20 A cont, 60 A peak (load-dependent).		1	Driver overhead <0.5 W; load power depends on supply (e.g., 12 V × 20 A ≈ 240 W, 24 V × 20 A ≈ 480 W)
Cytron MDD20A	6-30V Motor Supply; logic accepts 1.5-15V HIGH	Per channel up to 20 A cont, 60 A peak (load-dependent).		2	Driver overhead <1 W; both channels at 24 V → up to ~960 W total (load-dependent)
Cytron MDD10A	6-30V Motor Supply; logic accepts 1.5-15V HIGH	Per channel up to 20 A cont, 60 A peak (load-dependent).		4	Driver overhead <<1 W. Load power depends on your motor voltage: e.g. 12 V × 10 A ≈ 120 W per ch (both ch → ~240 W cont); 24 V × 10 A ≈ 240 W per ch (both ch → ~480 W cont).
N-MOS FET Trigger Module	Input Trigger 3.3-20V; load 5-36V	15 A continuous (up to 30 A with cooling); module self-consumption is small vs. load.		1	Overhead <0.5 W; max switched load up to ~540 W @ 36 V (derate for heat)
Relay SLD-24VDC-1C	24 VDC coil (operates ~19.2–28.8 V)	Coil power ≈ 1.6–1.8 W → ~67–75 mA @ 24 V; contacts NO 40 A / NC 30 A @14 VDC.		1	~1.6–1.8 W (coil)
Relay JQX-105F-4	5–110 VDC, 12–277 VAC variants	Coil power 0.9 W (DC) / 2 VA (AC) (e.g., at 12 VDC ≈75 mA); contacts up to 30 A.		1	~0.9 W (coil)
DC-DC Converter 300W /20 A	6–40 V in	Up to 20 A out (15 A recommended), typical eff. 95–97%; output 1.2–36 V. Input current depends on load/efficiency.		2	Up to 300 W output (plan ≤240 W cont. unless well-cooled)

ESP32-C3-Mini	3.0–3.6 V	Peak TX currents (from datasheet): Wi-Fi 802.11b ~335 mA, g/n ~208–181 mA; RX ~94 mA; modem-sleep down to ~17–37 mA depending on clock.		1	~0.2–0.4 W (peaks ~1.1 W worst-case TX burst)
LoRa SX1278	run at 3.3 V	TX ~120 mA @ +20 dBm, RX ~10–12 mA, standby/sleep much lower.	LoRa demod works at negative SNR; typical –20 dB at SF12	1	~0.04 W RX / ~0.40 W TX
4-Way Relay Module	12 VDC coil	~30 mA per relay (coil 400 Ω , ~0.36–0.45 W) → ~120 mA if all four ON, + a few mA for the board. Contacts typically 10 A.		1	~0.36 W per relay; all 4 ON → ~1.4–1.8 W total
Arduino Mega 2560 R3	5 V logic; input 7–12 V (6–20 V limit)	Board itself typically ~50–75 mA (no heavy I/O). 3.3 V pin max 50 mA		1	~0.25–0.38 W (board only)
VTX	12VDC	~400 mA (Power: 1.5 W RF out; 12 CH.)		1	~4.8 W
Arduino Mega 2560 Pro Mini	5 V logic	≈220 mA		1	~1.1 W
XL4015 5A Buck Module	3.2–40 V in, 1.5–35 V out	5A		1	Capability examples: 5 V×3 A ≈ 15 W; 12 V×2 A ≈ 24 W (keep the chip cool)
Arduino Pro Mini 5V 16 MHz	5 V (RAW pin accepts ~5–12 V via onboard regulator)	Active current typically ~15–20 mA at 5 V/16 MHz (≈17 mA measured; + a few mA for power LED/regulator).		1	~0.08–0.10 W
Camera Module	DC 12 V	Typical draw 50–100 mA	≥50 dB	4	~0.6–1.2 W (IR OFF)

Pack Energy & Usable Fraction for NiMH

- ➔ Nominal Energy: $24V \times 150 \text{ Ah} = 3,600\text{Wh}$
- ➔ Usable (NiMH):
 - ~70-80% (Voltage Sag + Higher Losses)
 - Use 75% as sane middle -> $\sim 2,700\text{Wh}$ usable

Average Load

- Motors: $10A \text{ total @ } 24V \sim 240W$ -> +10% overhead $\sim 264W$
- Electronics: $\sim 18W$ peak ($16W$ typical)
- Total avg: $\sim 282W$

Approximate Runtime (NiMH)

- Runtime $\approx$ usable Wh / avg W
- With 75% usable -> $2,700 / 282 \sim 9.6$ hours
- Range for different usable fractions:
- **70%:** $2,520 / 282 \sim 8.9h$
- **80%:** $2,880 / 282 \sim 10.2h$

6-Week Integrated Plan (ROV + Control Station)

Week 1 — Power Backbones & Safety (32–42 h)

ROV (18–26 h)

- 24 V NiMH main bus, main fuse, busbars, branch fuses, labels (6–8 h)
- 24→12 V buck (video/RF), 24→5 V buck (logic), XL4015 sub-rail; dummy-load tests (5–7 h)
- SLD 24 V coil, MOSFET trigger bench checks, flyback, ground layout (2–3 h)
- Voltage-sag tests @ 5–20 A pulses; log drop & recovery (3–4 h)
- T&E buffer: wiring rework/ground loop chasing (2–4 h)

Controller (14–16 h)

- Enclosure layout; battery bay; 24→12 V (VRX), 24→5 V (MCU/LoRa) rails (4–5 h)
- Joysticks, toggles, pots wiring; pinout doc (4–5 h)
- Mount 1.5 GHz VRX, SMA pigtails, isolated 12 V feed (3–4 h)
- T&E buffer: 2 h

Milestone M1: Both platforms power cleanly; no resets on load pulses; controller shows bench video.

Week 2 — Propulsion & I/O Bring-up (34–44 h)

ROV (20–28 h)

- Spin each driver: MD20A, MDD20A×2, MDD10A×4 with one motor each (4–6 h)
- Stage dual/quad channels; current sharing & thermals (5–7 h)
- PWM freq/decay tuning per driver family (4–5 h)
- Fault tests: stall/brownout/thermal throttle (3–4 h)
- T&E buffer: 4–6 h

Controller (14–16 h)

- MCU (ESP32-C3 or Mega) reads sticks/switches; sanity serial UI (4–5 h)
- Mechanical: grips, strap, serviceability (2–3 h)
- Power noise check with VRX on (3–4 h)
- T&E buffer: 2–3 h

Milestone M2: All motor channels spin reliably; controller inputs read stable.

Week 3 — Radio Links & Baseline Control (30–40 h)

ROV (14–18 h)

- Control stack scaffolding on Mega: task loop, states, mixer stub (5–6 h)
- LoRa SX1278 on ROV: IDs, CRC, watchdog, basic telemetry (4–5 h)
- I/O map: relays, E-stop, arming (2–3 h)
- T&E buffer: 2–4 h

Controller (16–22 h)

- LoRa protocol v1: RC channels, heartbeat/ACK/timeout (5–6 h)
- Telemetry downlink (Vbat, RSSI/SNR, temps); SD/flash logs (4–5 h)
- Config page (serial CLI or tiny web on ESP32-C3), channel map/rates (3–4 h)
- T&E buffer: 3–5 h

Integration Test I1 (2–3 h):

Bench: sticks → motors via LoRa; loss/reconnect within spec; log RSSI/SNR.

Milestone M3: Closed-loop RC control over LoRa on the bench with telemetry.

Week 4 — Video/RF Integration & EMI Hardening (28–38 h)

ROV (12–18 h)

- VTX + 4× cameras on 12 V rail (IR control if needed); ferrites/LC on VTX feed (5–7 h)
- Ground antenna workflow (patch/Yagi vs omni) + quick shoreline check (3–4 h)
- Noise hunts: star-ground, harness reroute, VRX/VTX isolation from motor spikes (3–4 h)
- T&E buffer: 1–3 h

Controller (16–20 h)

- VRX to monitor/goggles; OSD/OLED for RSSI/Vbat/mode (6–8 h)
- Data recorder UX: file rotation, “mark event” button (3–4 h)
- Antenna mounts: quick-swap directional/omni; aiming card (3–4 h)
- T&E buffer: 2–4 h

Integration Test I2 (2–3 h):

Range-of-control walk test; verify video stays clean during TX bursts; log EMI incidents.

Milestone M4: Usable operator station: clean video + stable control at short range.

Week 5 — Features, Failsafes & Tuning (34–44 h)

ROV (18–24 h)

- Mixer modes (manual/eco/cruise), slew limits/ramps (6–8 h)
- Failsafes (low RSSI, link loss, undervolt); idle/return-logic (4–5 h)
- Telemetry expansion (rail volts/temps/current snapshots) (4–5 h)
- T&E buffer: 3–6 h

Controller (16–20 h)

- Profiles (normal/eco/manual), expo/deadband per profile (4–5 h)
- Low-voltage alarms/buzzer, arming/SAFE latching with LEDs (4–5 h)
- Long-range rehearsal plan; pre-field checklist (3–4 h)
- T&E buffer: 3–6 h

Integration Test I3 (3–4 h):

Tethered or controlled water test: drive + video + telemetry; provoke failsafes; confirm recovery.

Milestone M5: Operator can drive, see telemetry, and pass failsafe tests.

Week 6 — Soak, Range, Hardening & Docs (32–42 h)

ROV (16–22 h)

- 2–4 h soak at ~13–14 A avg; temp & voltage logs (6–8 h)
- Waterproofing, strain relief, potting; thermal refinements (5–7 h)
- Final antenna placement & shoreline range test (3–4 h)

Controller (10–14 h)

- Ergonomics: balance, handles, glare mitigations (3–4 h)
- Spare fuse/parts kit, service procedure (2–3 h)
- User docs: wiring, radio config, field checklist (3–4 h)

Integration Test I4 (3–6 h):

Over-water mission rehearsal: stepped range, antenna swaps, record RSSI/SNR vs distance; verify logs.

Milestone M6 (Final): 2–4 h field run without faults; docs complete; kit packed.

#DOESN'T INCLUDE WORK HOURS FOR 3D MODELLING AND PRINTING

COST

ITEM	QTY	UNIT PRICE (RM)	REQUIREMENT
Cytron MD20A (1-ch 20A)	1	90	
Cytron MDD20A (2-ch 20A)	2	150	
Cytron MDD10A (2-ch 10A)	4	100	
N-MOSFET trigger module (15–30 A style)	1	10	
Relay SLD-24VDC-1C	1	8	
Relay JQX-105F-4	1	30	
DC-DC 300 W / 20 A buck	2	35	Need to Recheck
ESP32-C3 mini / SuperMini	1	20	Need to Recheck
LoRa SX1278 (433 MHz)	1	40	Need to Recheck RF & Antenna
4-way relay module 12 V	1	25	
Arduino Mega 2560 R3	1	239	
VTX + VRX kit (1.5 GHz 1500 mW)	1	590	Need to Recheck RF & Antenna current VTX & VRX having problem other kit come at price RM1k+
Arduino Mega 2560 Pro Mini	1	70	
XL4015 5 A buck module	1	12	Need to Recheck
Arduino Pro Mini 5 V 16 MHz	1	35	
12 V camera module	4	35	Need to Recheck

Doesn't Include Wire and Other Instrument such as Capacitor/Resistor and else.